Francisco Calvillo Marmaneu

CURRICULUM VITAE

Education

• PhD in Mathematics: Probabilistic Methods for Structure Recovery in Random Discrete Models 2023 - 2026

Sorbonne University & Pompeu Fabra University
Advisors: Nicolas Broutin, Gábor Lugosi

• Paris-Saclay University 2021-2022

Paris-Saclay

- 2021-2022: Master's degree (Year 2), Mathematics of Randomness(Limit Theorems and Applications, Random Graphs, Ergodic Theory, Brownian Motion and Stochastic Calculus, Random Matrices, Topological Data Analysis, Solvable Models in Probability, Local Times and Excursion Theory)

• École Normale Supérieure (Ulm) 2019-2023

Paris, 5th arrondissement

- 2019-2020: Bachelor's degree in Mathematics (Integration and Probability, Topology and Differential Calculus, Complex Analysis, Group Theory, Quantum Mechanics, Statistical Physics)
- 2020-2021: Master's degree (Year 1) in Mathematics (Stochastic Processes, Dynamical Systems, Differential Geometry, Functional Analysis, Algebraic Topology, Statistics)

• Lycée Saint-Louis: classe préparatoire 2017-2019

Paris, 6th arrondissement

- MPSI (Mathematics, Physics, Engineering) 2017-2018, specialization in Computer Science.
- MP* (Mathematics, Physics) 2018-2019, specialization in Computer Science.

• Lycée Français de Valence 2004-2017

Valencia, Spain

- Scientific Baccalaureate, specialized in Mathematics (With Highest Honors)

Talks and Invitations

• Invited participant. 6th Workshop on Geometric, Algebraic and Probabilistic Combinatorics, Poblet, ES. Jun. 2025

• Invited talk at Fernuni Math and CS department Seminar, Unidistance, Brig, CH. Jun. 2025

• Invited participant. Bellairs workshop on Probability and Combinatorics, McGill Bellairs institute, Holetown, BB. 2024, 2025

• Contributed talk at Oxford Probability Workshop: Random Discrete Structures, University of Oxford, Oxford, UK. Mar. 2025

• Invited participant. Workshop on Randomness and Learning on Networks, Combinatorics, IMPA, Rio de Janeiro, BR. Aug. 2024

• Invited talk at LIMDA seminar, Universidad Politécnica de Catalunya, Barcelona, ES. Jun. 2024

• Invited talk at IME, Universidade de São Paulo, São Paulo, BR. May 2024

Publications

[1] Simon Briend, Francisco Calvillo, and Gábor Lugosi. Archaeology of random recursive dags and cooper-frieze random networks. *Combinatorics, Probability and Computing*, 32(6):859–873, 2023.

[2] Francisco Calvillo, Luc Devroye, and Gábor Lugosi. Subtractive random forests with two choices. *Methodology and Computing in Applied Probability*, 28, 05 2026.

[3] Sofiya Burova, Francisco Calvillo, Gábor Lugosi, and Piotr Zwiernik. Testing properties of trees in graphical models with covariance queries, 2026. Submitted for publication 2026 https://arxiv.org/abs/2605.15996.

[4] Sofiya Burova and Francisco Calvillo. Temporal inhomogeneous random graphs. *Submitted for publication*, 2026.

Research experience

• Research visit (RandNET program) Oct. 2023
McGill University

• Predoctoral Internship: subtractive random forests, squarish k-d trees Sept. 2022 - Aug. 2023
Pompeu Fabra University, supervisors: Nicolas Broutin and Gábor Lugosi.

• Master's Internship: root-finding algorithms on random recursive DAG's and linear preferential attachment graphs May 2022 - July 2023
Sorbonne Université, supervisor: Nicolas Broutin.

• Bachelor's Internship: archeology of the simplified Cooper-Frieze model Feb. 2021 - June 2021
Pompeu Fabra University, supervisor: Gábor Lugosi.

• Bachelor's Thesis: Hausdorff dimension of p-adic subrings Feb. 2020 - June 2020
École Normale Supérieure, supervisor: Timothée Bénard

Teaching

• Teaching assistant for the Calculus and Linear Algebra brush up Sep. 2025
Pompeu Fabra University

Technical Skills

Programming: Python, OCaml, LaTeX

Languages

Spanish (Native); French (Native; B2 certified); English (Fluent; B2 certified)